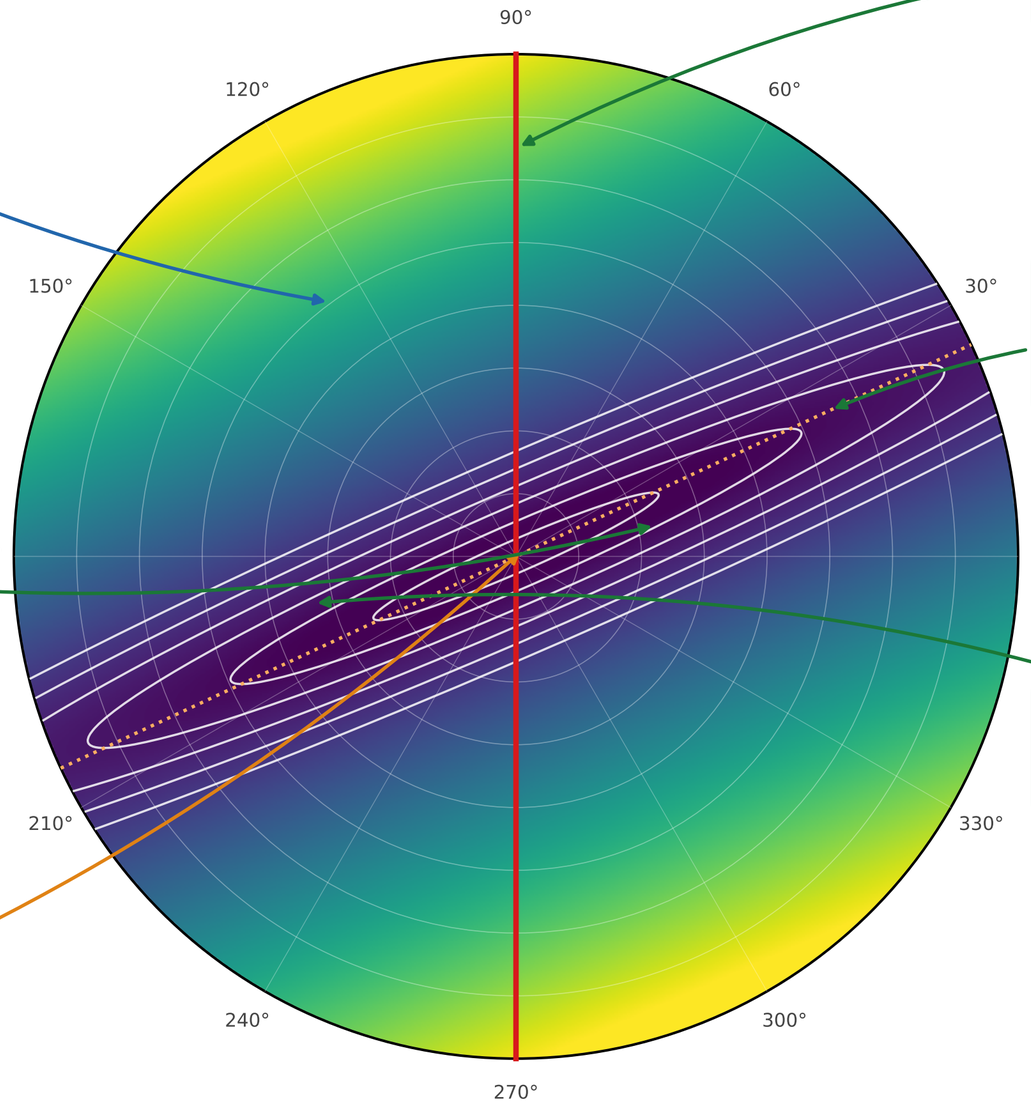
Stage 2 — Approach to degeneracy

Same norm conic with the trace pushed toward the Hasse edge: a = 4.3 (illustrative sweep value), a² = 18.49 < 4q = 20. The lobes stretch along the future null direction.

nested rings = weight strata |z| = q^(−σ)
= attenuation classes; purity = one ring
= distortionless (÷ q^½ → lossless)
[Restatement for Frobenius / Analogy for ζ]

y-axis = critical line Re s = ½:
the unitary spectrum of scaling
U_λ f(x) = λ^½ f(λx) on half-densities
(|λ̄^(½−s)| = 1 ∀λ ⇔ Re s = ½; TAA Lemma 153 §5)
[Theorem; frame axis = Analogy]



lobes elongate along m = (a/2)·n:
discriminant a² − 4q → 0⁻; eigenvalues F, V =
(a ± i√(4q − a²))/2
slide toward each other on the circle |z| = √q
[Theorem]

Krein one-liner: ⟨Fx,Fy⟩ = q⟨x,y⟩ ⇒ |α| = √q
or v null —
still definite here; MP4 is the only
positivity input
[Theorem]

Legendre support window still 2π
(ellipse: every direction has a tangent);
it jumps 2π → 0 at det = 0 (TAA Lemma 153)
[Theorem]

origin ↔ point at infinity = the cusp:
constant term $a_0(y) = y^s + \varphi(s)\cdot y^{(1-s)}$,
|φ(½+it)| = 1 (TAA Lemma 153 NUM ≤ 2e−31)
[Analogy for the identification]

Status tags: [Theorem] [Restatement] [Analogy] [Numerical] [Cited] — see legend.png. RH is not proved. No null rays at 45° (per J. W. McGreevy); lobes are elliptical conic sections, not elliptic curves.

After J. W. McGreevy's animation 'Flat Potential Oscillations' (Aug 2025, x.com/McGreevyMD/status/1951474097990279331). Re-rendered by Grok Bot from the analysed geometry; not J. W. McGreevy's code.